

FitFlow User's Guide

UMGC CMSC 495

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Introduction

The team behind the **FitFlow** Interactive Workout Assistant is proud to provide the latest in fitness tracking software. Our software allows the User the ability to build, track, and record their workouts and progress. The goal of this application is to improve the health of people globally by increasing user exercise awareness and enhancing the workout experience.

FitFlow Fitness Tracker

The program is a Java-based application that can be run on the user's home desktop or laptop computer and should be platform (operating system/OS) independent. Users will register an account via a password system and be able to track their fitness progress over time. By registering an account, users will have privacy and security over sensitive data, and access to workout/exercise statistics.

System Features:

- Java based application utilizing the latest JVM allowing for platform independence.

Overview of the Software

Purpose of the User Guide

The purpose of the **User Guide** is to inform the reader and User of the application how to successfully navigate and make the best use of **FitFlow**.



Figure 1. (University of Maryland Global Campus, 2020).

Target Audience

“**FitFlow** is for everyone!”

The **Target Demographic** or **audience** of the Team behind **FitFlow** are primarily people interested in increasing their fitness, who have trouble keeping track of their target fitness goals, and could benefit from our application’s assistance.

Getting Started

System Requirements

In order to use **FitFlow**, Users must have an electronic device with an **operating system (OS)** with an **Integrated Development Environment (IDE)** installed that is capable of running **Java JDK 26** or higher, and they must have **JavaFX** configured in their IDE.

Installation Instructions

To install the program, you should visit [FitFlow’s GitHub page](#), download the code repository (repo) from the webpage, unzip the file, open the project in their preferred Java-based IDE, and run;

```
src/main/java/app/Main.java
```

Users should ensure that *JavaFX is configured* in their IDE before running the program, or they might experience User-Interface (UI) issues.

User Interface Overview

Creating A User Account

Upon successful launch of the program, you will first be met with the **Login Screen**;

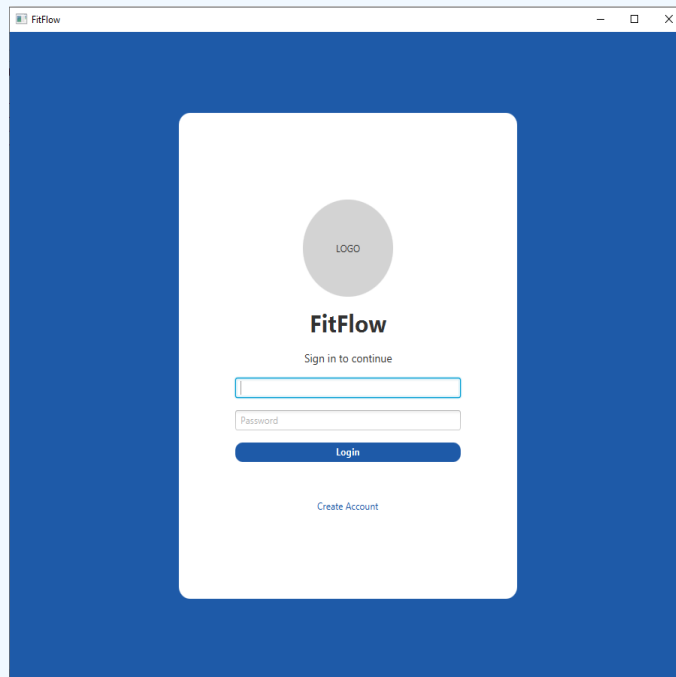


Figure 2. Screenshot of **FitFlow's** Login Screen.

The first time you use the **FitFlow** Fitness Tracker, you will need to register a **User Account** with the program. This will **password protect** your data and provide privacy while you undergo your fitness transformation. A User Account can be created by clicking the “**Create Account**” Button on the Login Screen.

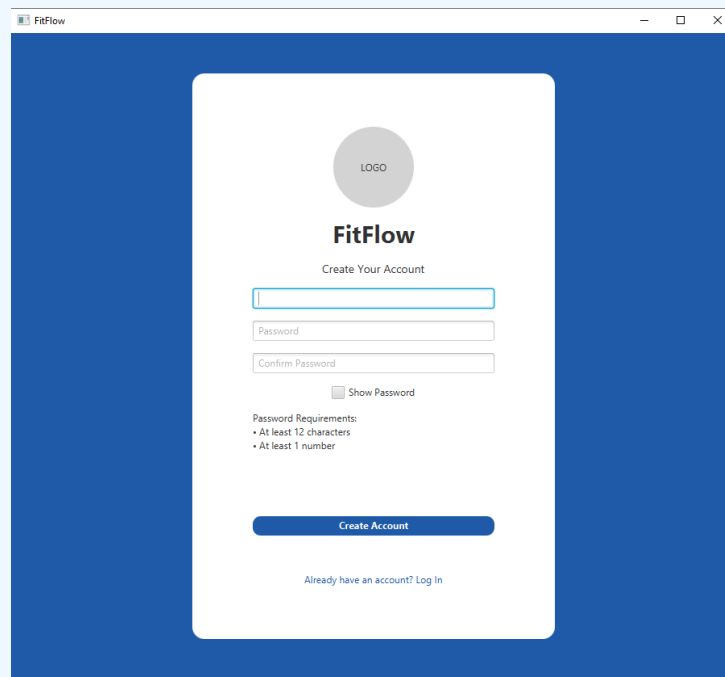


Figure 3. Screenshot of **FitFlow’s Registration** screen.

To successfully create an account, you will be required to select and type in a unique **username**, and associated **password**. Each character entered into the password fields are hidden behind the “●” character, but still recorded by the system to ensure user privacy during account creation. If this is not a problem however, you may select the “**Show Password**” box to display the password for easier use during account creation.

Additionally, by typing in the desired password twice, **FitFlow** ensures the User is aware of any mistakes they might have made during the account creation process.

Users should ensure that their desired password is unique, twelve characters long or longer, includes at least one number, and that the typed in password in both password fields match.

The screenshot shows a web form for creating a new account on the FitFlow platform. At the top, there is a circular logo placeholder labeled "LOGO". Below it, the brand name "FitFlow" is displayed in a large, bold, black font. Underneath the brand name, the text "Create Your Account" is centered. The form contains three input fields: the first for a username, which has "Tim3" entered; the second for a password, which has "asdfghjkl123" entered; and the third for a confirmation password, which also has "asdfghjkl123" entered. Below the password fields, there is a checkbox labeled "Show Password" which is currently checked. Underneath the checkbox, the text "Password Requirements:" is followed by a bulleted list: "• At least 12 characters" and "• At least 1 number". Below the list, two green status messages are displayed: "Password meets requirements." and "Passwords match.". At the bottom of the form, there is a prominent blue button with the text "Create Account". Below the button, there is a link that says "Already have an account? Log In".

LOGO

FitFlow

Create Your Account

Tim3

asdfghjkl123

asdfghjkl123

☒ Show Password

Password Requirements:

- At least 12 characters
- At least 1 number

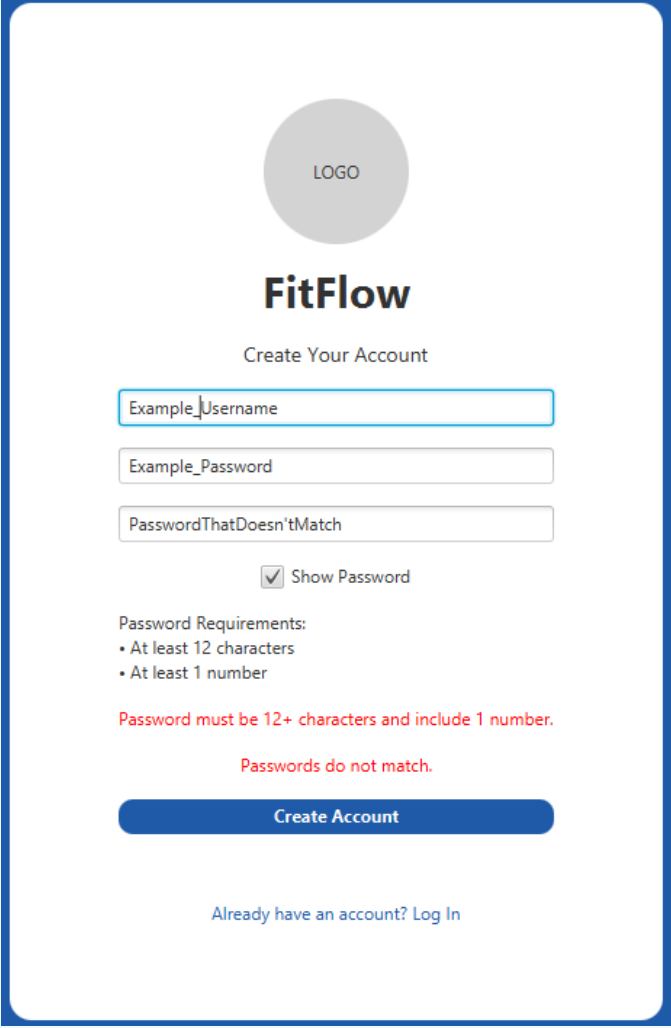
Password meets requirements.

Passwords match.

Create Account

Already have an account? [Log In](#)

Figure 4. Screenshot demonstrating passwords matching.



The screenshot displays the 'FitFlow' account creation interface. At the top, there is a circular logo placeholder labeled 'LOGO'. Below it, the 'FitFlow' logo is prominently displayed. The heading 'Create Your Account' is centered above the input fields. There are three input fields: the first contains 'Example_Username', the second contains 'Example_Password', and the third contains 'PasswordThatDoesn'tMatch'. A checkbox labeled 'Show Password' is checked. Below the fields, the 'Password Requirements' are listed: 'At least 12 characters' and 'At least 1 number'. A red error message states 'Password must be 12+ characters and include 1 number.' Another red error message below it says 'Passwords do not match.' A blue 'Create Account' button is positioned below the errors. At the bottom, a link reads 'Already have an account? Log In'.

Figure 5. Screenshot demonstrating unsuccessful account creation.

Once this is done, you should click **‘Create Account’** to register an account with **FitFlow**. This ensures all information like user-built workouts, or important health data are saved to the user’s profile for later use, saving the user time.

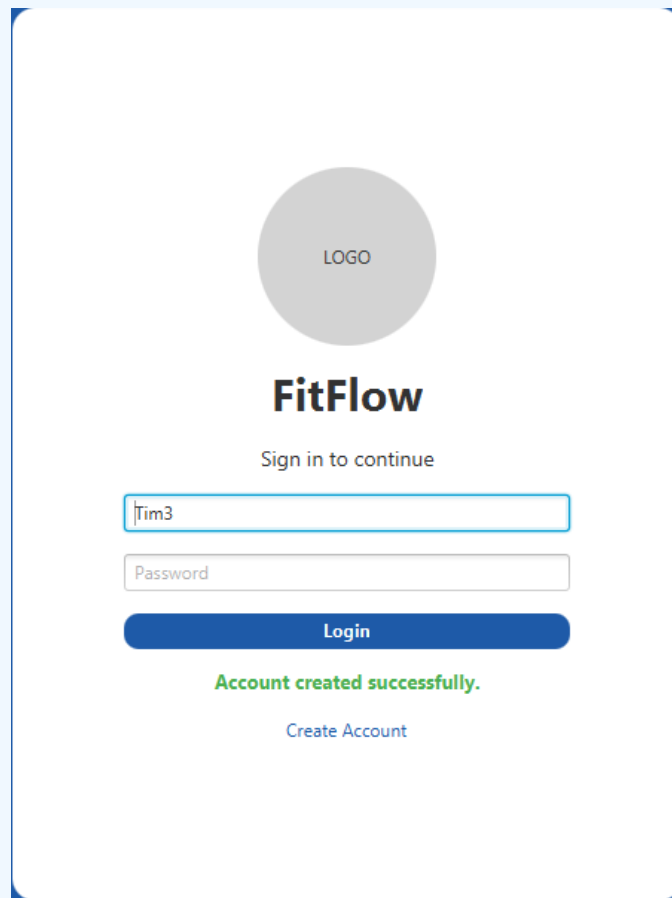


Figure 6. Screenshot demonstrating successful account creation.

Upon successful account creation, **FitFlow** will direct you back to the **Login Screen**, to log in into the program.

Software & Main Feature Step-by-Step Instructions

Navigation

Upon successfully logging into the program, **FitFlow** will direct you to the **Dashboard** screen, from which the User can easily navigate **FitFlow**;

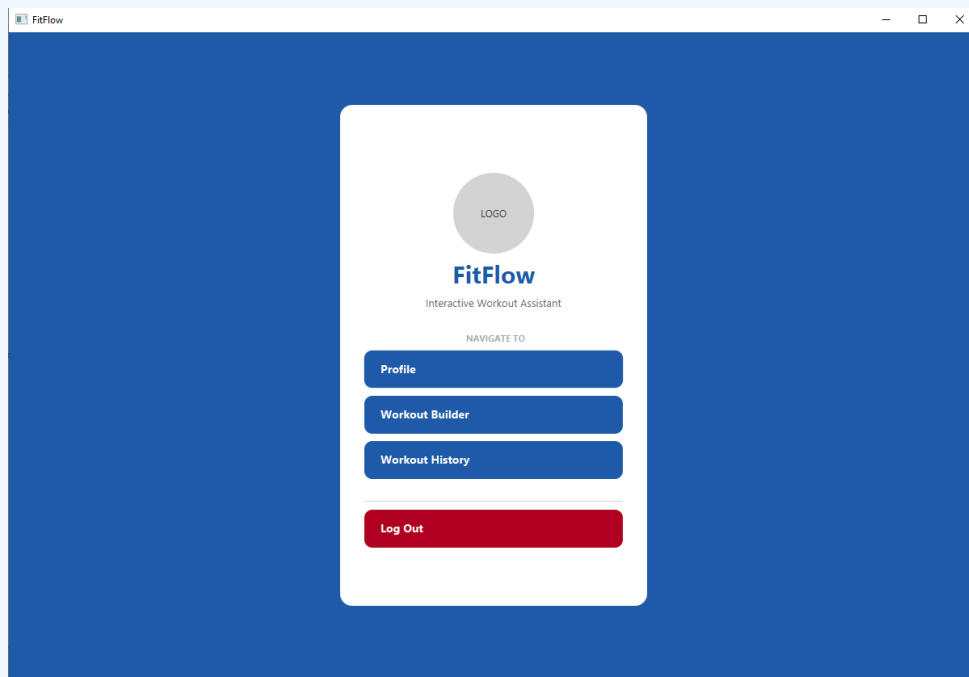
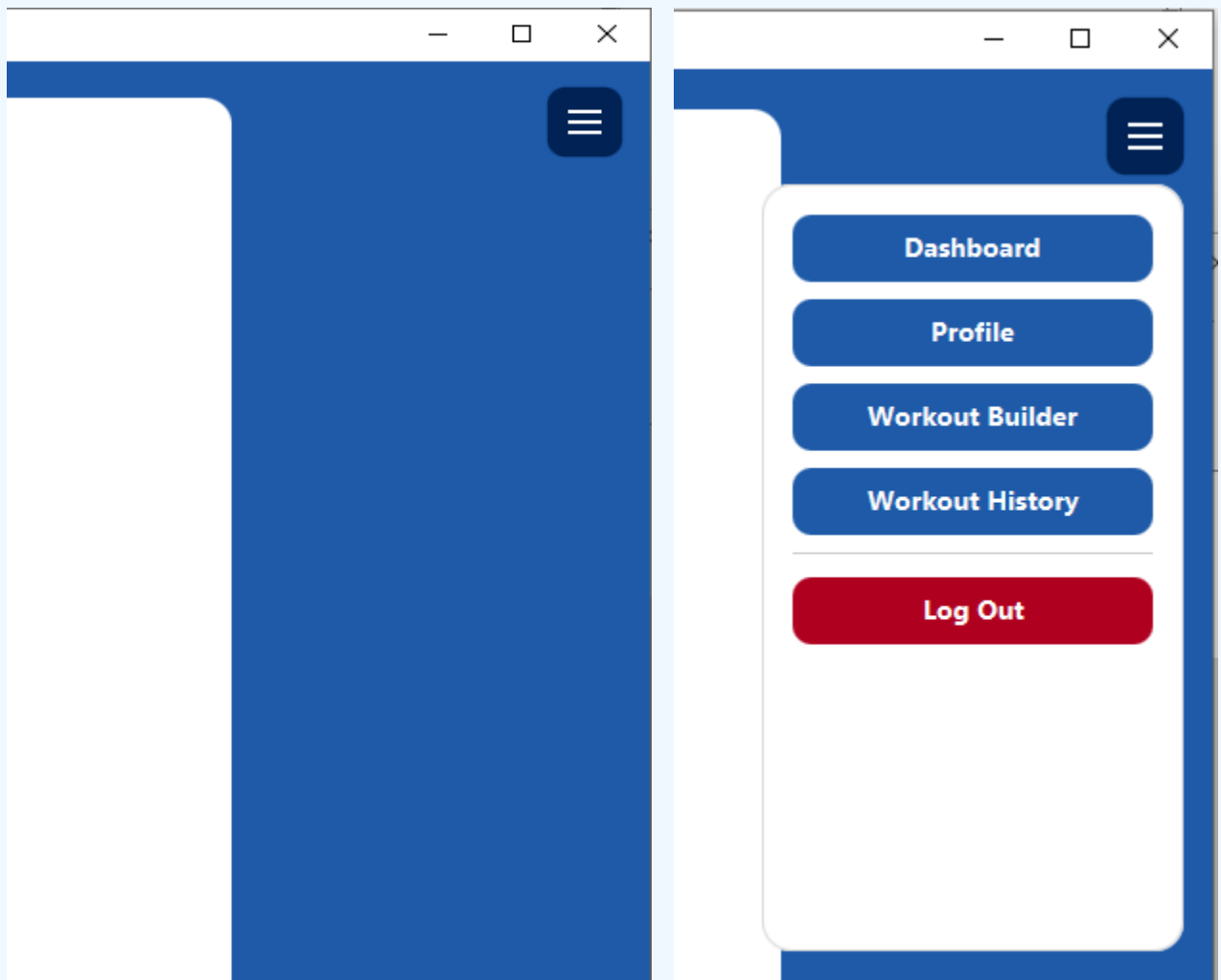


Figure 7. Screenshot of the **Dashboard Screen**.

From here, you may click the **Profile** button to go to the **Profile Screen**, the **Workout Builder** button to be directed to the **Workout Builder**, or the **Workout History** button to be directed to the screen displaying your **workout history**.

Alternatively, you may click the **Log Out** button to be successfully logged out of the program.



Figures 8 & 9. Screenshot of the **Navigation Menu** or the “hamburger menu”, before (left) and after (right) clicking the icon.

After clicking a button on **Dashboard screen** and being directed to the corresponding screen, you may return to the Dashboard Screen or choose a specific screen by clicking on the **Navigation Menu** icon or the “hamburger menu” at any time.

Doing so will bring up a menu of buttons. From here, you may click the **Profile** button to go to the **Profile Screen**, the **Workout Builder** button to be directed to the **Workout Builder**, or the **Workout History** button to be directed to the screen displaying your **workout history**.

Alternatively, you may click the **Log Out** button to be successfully logged out of the program.

Utilizing the Workout Builder

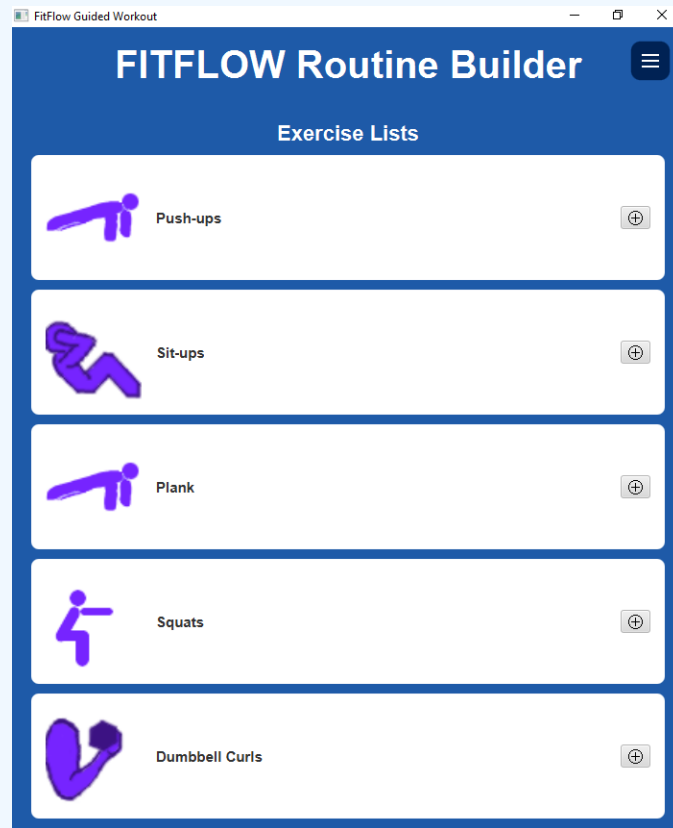


Figure 10. Screenshot of the **FitFlow Routine Builder** landing page.

After being directed to the **FitFlow Routine Builder** landing page, you may construct a workout by picking the desired exercises with a click of their associated “+” **button**. Doing so will populate your currently constructed workout to the right of the selection menu, as seen in **Figure 11**.

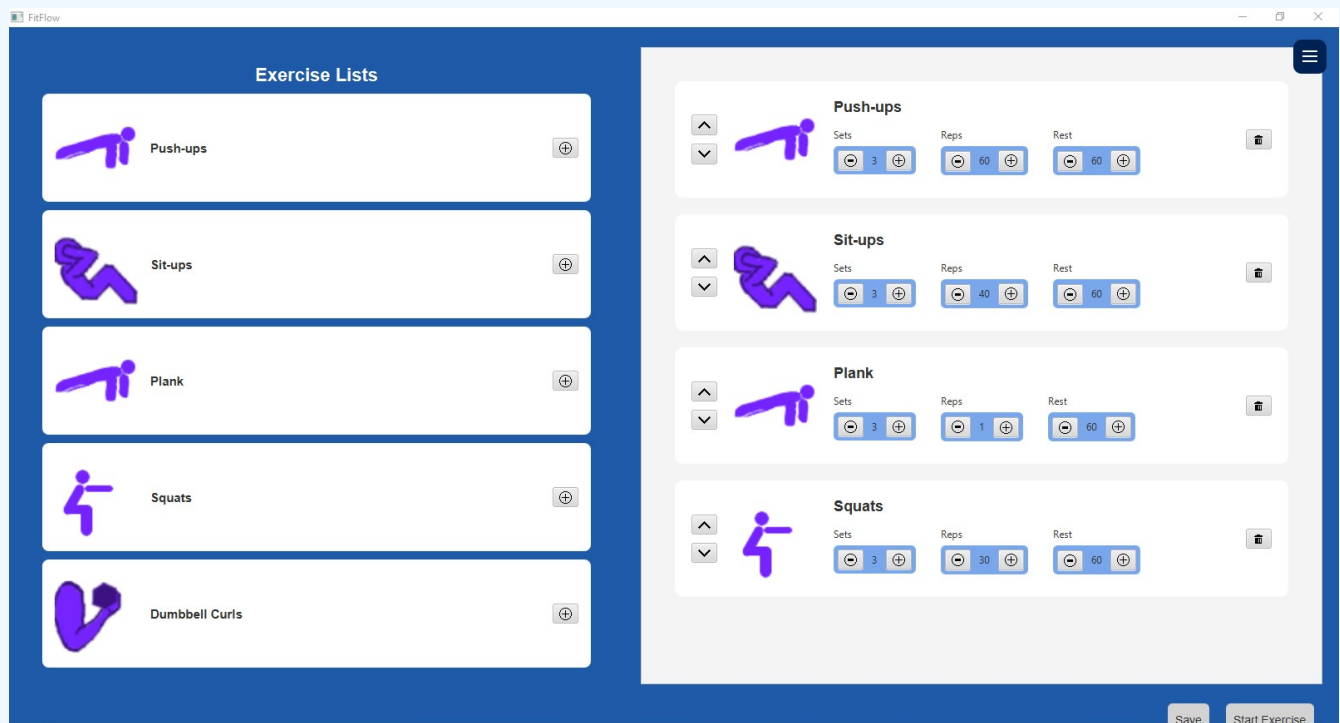


Figure 11. Screenshot of your average, standard workout constructed using **FitFlow**.

Once the selected exercises have been added to the menu, you may utilize the “▲” and “▼” **buttons** to arrange the exercises in your desired order.

Additionally, you may utilize the “–” and “+” **buttons** to increment or decrement the desired number of sets, reps, and rest-time given in seconds.

Furthermore, you may utilize the “🗑️” **button** to remove an exercise from the currently constructed workout.

Logging Workout Data

To save the workout, simply press the “**save**” **button**. Doing so will bring up the ‘Save Routine’ pop-up.

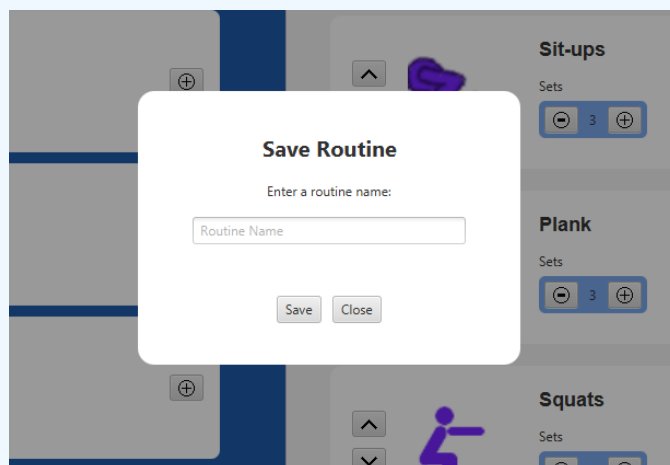


Figure 12. Screenshot of the ‘Save Routine’ pop-up.

After the Save-Routine pop-up appears, you may enter anything you like to name your workout. Once you’ve come up with a satisfactory name for your workout routine, you may save it to **FitFlow’s** system by pressing the **“Save” button**. Doing so will display in green text, *‘Workout routine saved successfully’*.

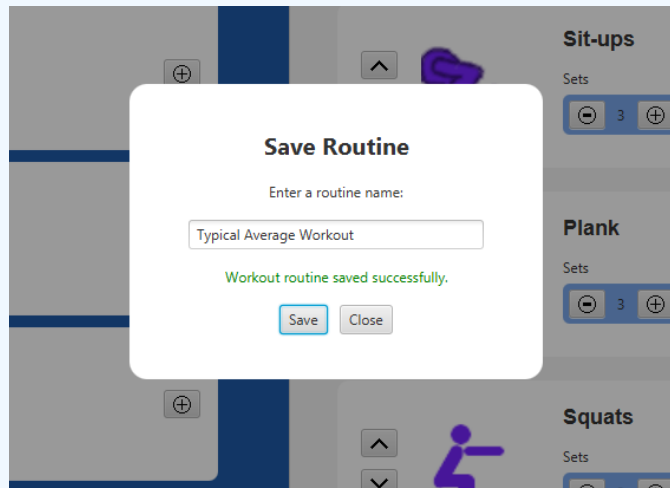


Figure 13. Screenshot demonstrating successfully saving a workout routine.

To continue, simply press the **“Close” button** to exit the **Save-Routine pop-up**. This will return you to the **Workout Builder** screen.

Guided Workout Mode:

From the **Workout Builder** screen, click the **“Start Exercise”** button to begin the constructed workout.



Figure 14. Screenshot of the **FitFlow Guided Workout** screen.

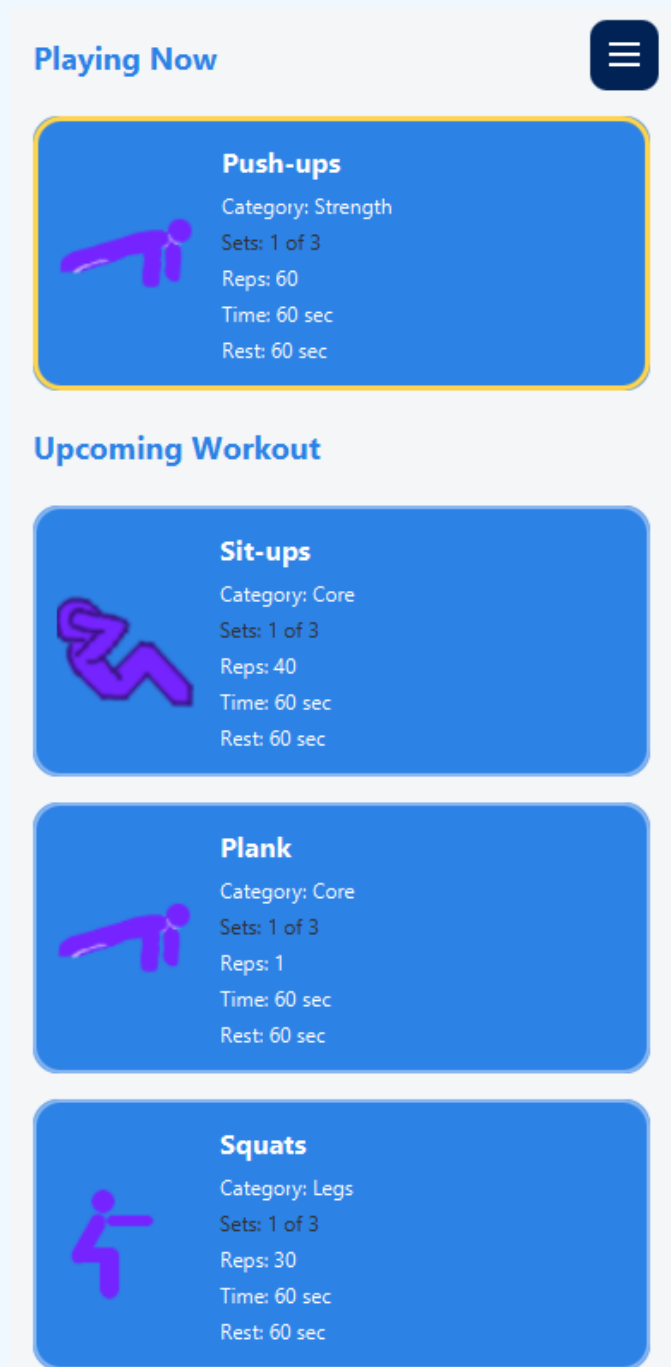


Figure 15. Screenshot of **FitFlow's** Workout Queue.

Workout Queue:

FitFlow's Workout Queue will display on the right-hand side of the screen, displaying useful information like exercise type, the category of muscle-group that the exercise works, the selected amount of *Sets*, *Reps*, *Set-Time*, & *Rest-Time* to the User.

Under, **“Playing Now”**, **FitFlow's Workout Queue** will display to you *the current exercise playing* in the constructed workout.

Under, **“Upcoming Workout”**, **FitFlow's Workout Queue** will display to you *the upcoming exercises* in the constructed workout.

To **begin** the workout, click the “▶”/“**Play**” button. The **Set-Timer** will begin counting down in seconds for the desired set.



Figure 16. Screenshot showing **FitFlow's Set-Timer** countdown.

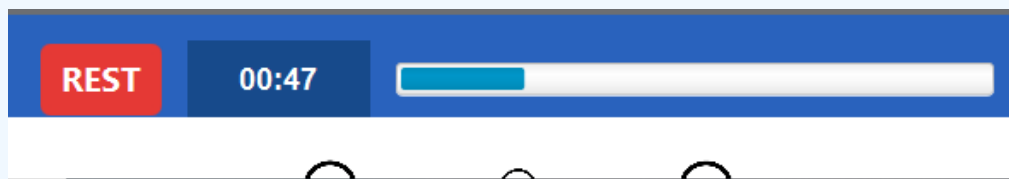


Figure 17. Screenshot showing **FitFlow's Rest-Timer** countdown.

To **pause** the workout, click the “⏏”/“**Pause**” button.

To **move on** to the next exercise, click the “⏭”/“**Skip**” button.

To **go back** to the last exercise, click the “⏮”/“**Back**” button.

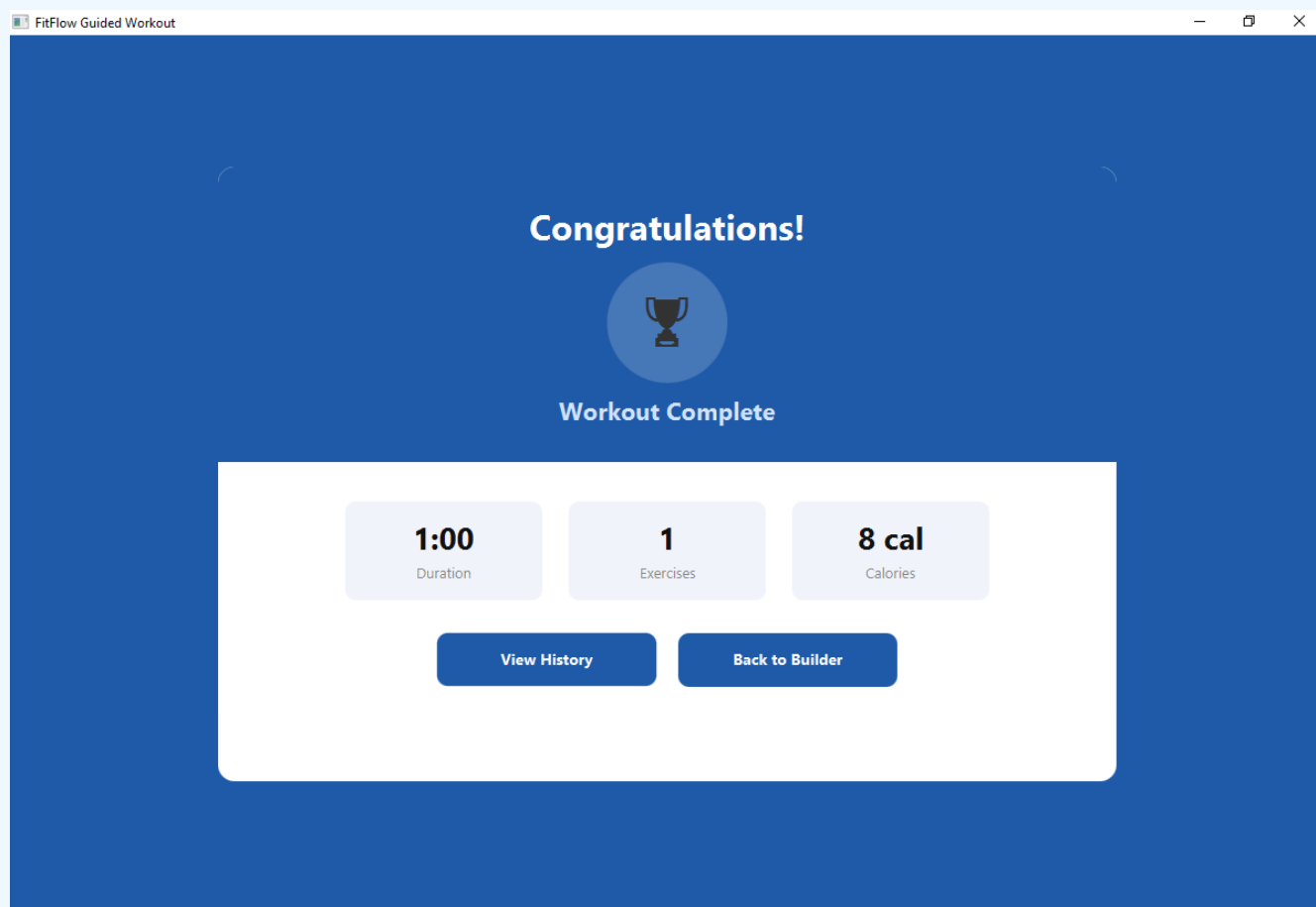


Figure 22. Screenshot of the Congratulations Screen.

Upon successful completion of your workout, you will be met with the **Congratulations Screen**, along with a summary of the workout's duration, exercise count, and estimated burned calorie count.

You may press, '**View History**' button to navigate to the, '**Workout History**' screen or press the, '**Back to Builder**' button to navigate back to the **Workout Builder** screen.

Workout History

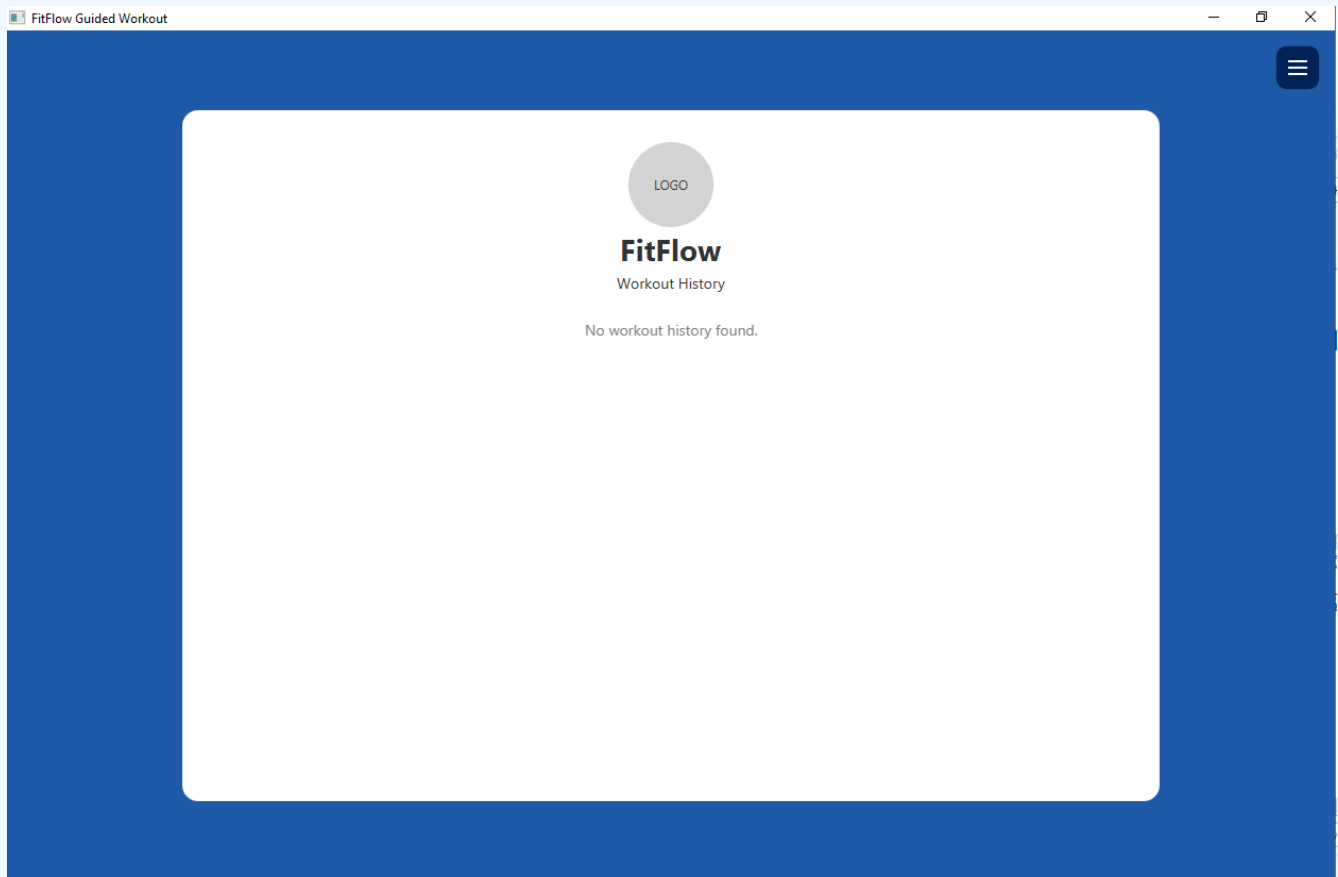


Figure 18. Screenshot depicting **FitFlow's Workout History** screen. Presently empty until the User has completed their first workout.

Upon the initial launch of the program, **FitFlow's Workout History** screen will be empty. However, after you have completed your first workout, **FitFlow** will record your workout, where you may view it by navigating to the **Workout History** screen.

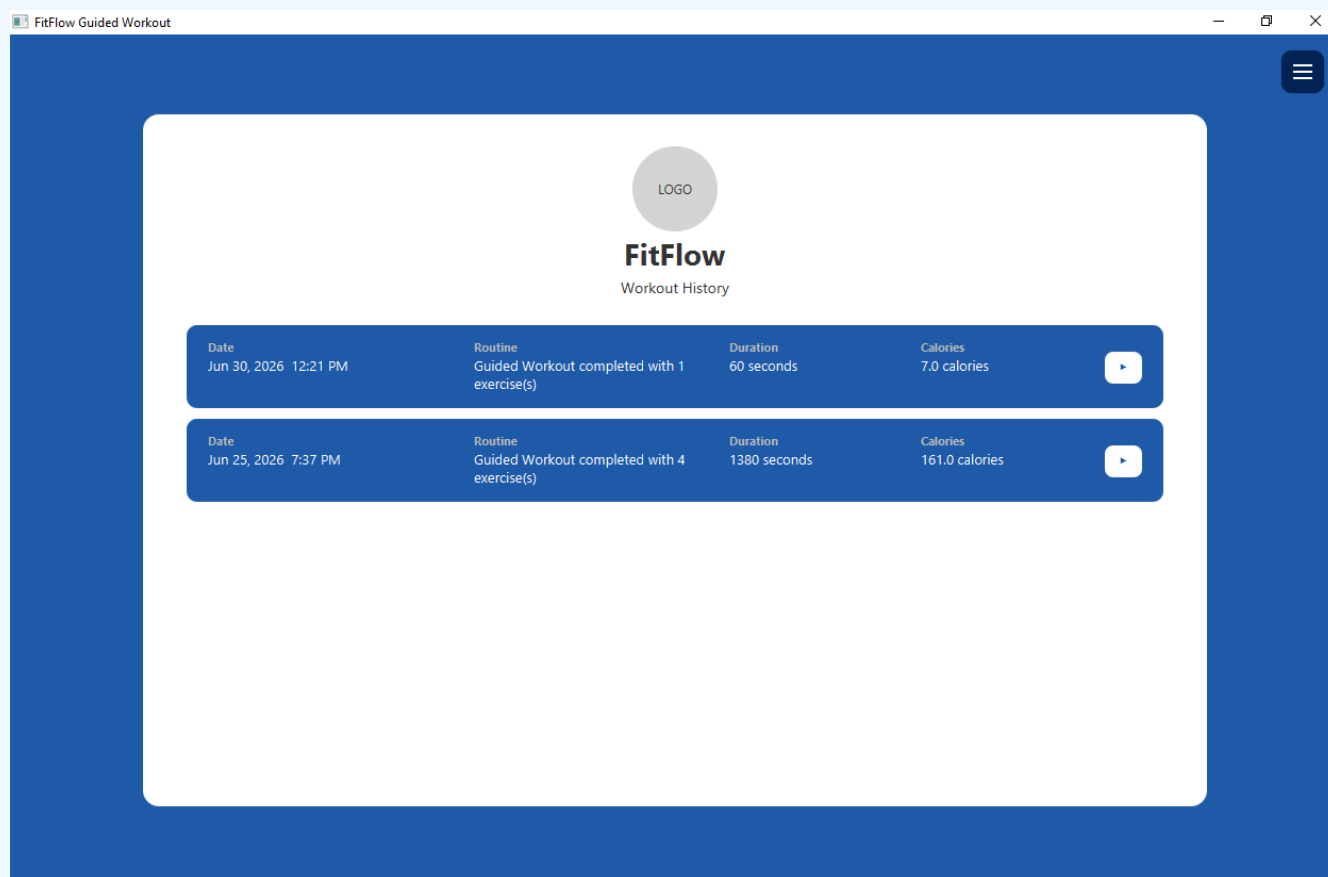


Figure 19. Screenshot depicting **FitFlow’s Workout History** screen, now populated with data from User workouts.

FitFlow records the **Date** (YYYY-MM-DD | hh:mm:ss), **Routine**, **Duration** (seconds), and estimated **Calories burned** for the User per each session.

You may press the “▶”/“**Play**” button to be redirected back to the **Guided Workout Mode** where your saved workout will be loaded and ready to go through again. Simply click the, ‘**Start Exercise**’ button to do your saved workout again.

Entering and Updating User Profile Information

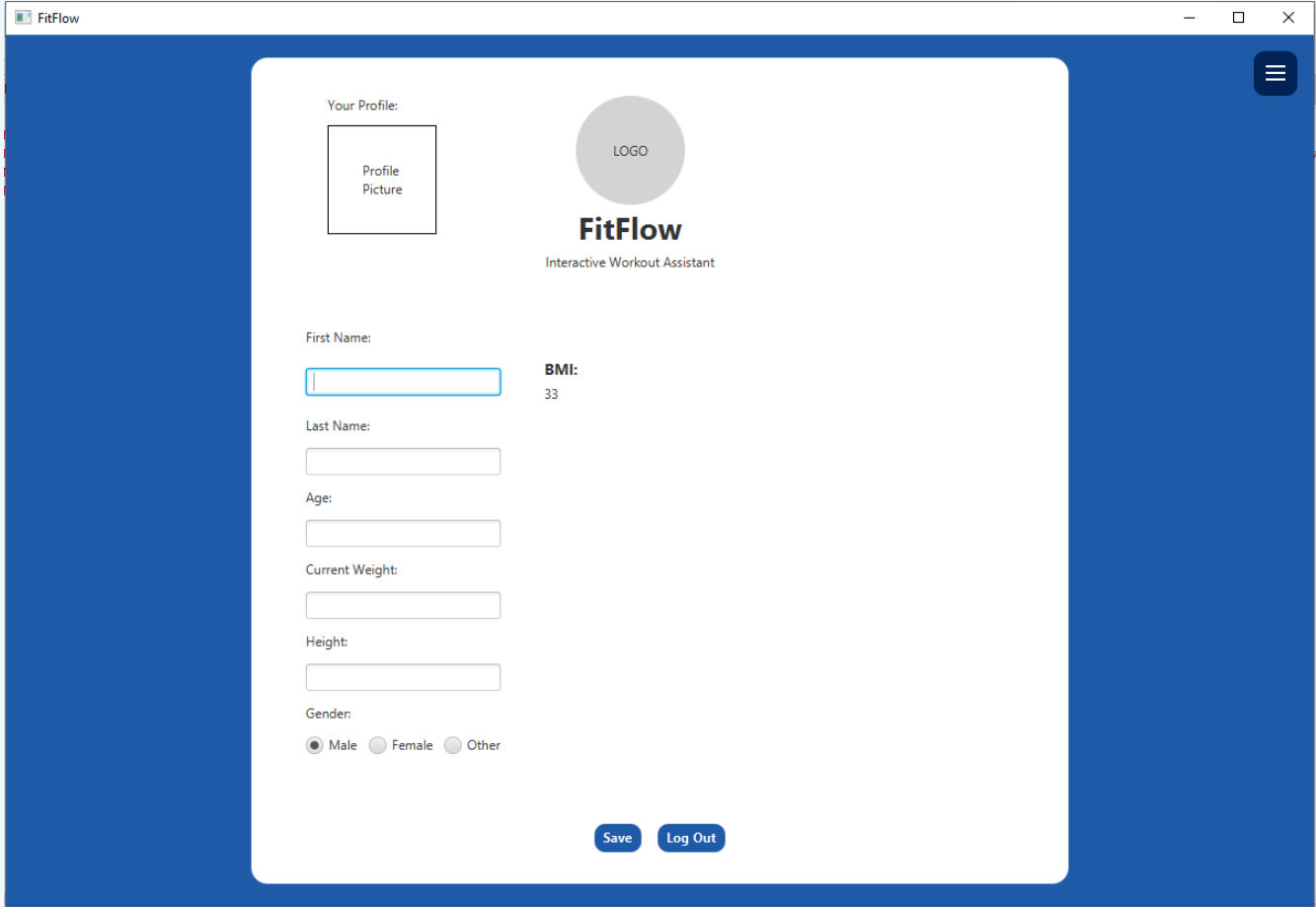
A screenshot of a web application window titled "FitFlow". The window has a blue sidebar on the left and a white main content area. In the top right of the sidebar is a hamburger menu icon. The main content area is titled "Your Profile:" and contains a "Profile Picture" placeholder box on the left and a circular "LOGO" placeholder on the right. Below the logo is the text "FitFlow" and "Interactive Workout Assistant". The form fields are arranged in two columns. The left column contains: "First Name:" with a text input field, "Last Name:" with a text input field, "Age:" with a text input field, "Current Weight:" with a text input field, "Height:" with a text input field, and "Gender:" with three radio buttons labeled "Male", "Female", and "Other". The right column contains: "BMI:" with the value "33" displayed below it. At the bottom of the form are two buttons: "Save" and "Log Out".

Figure 20. Screenshot of the **Profile Screen**.

On the **Profile Screen**, you can record and save information saved to a unique **User Profile**. **FitFlow** provides fields for you to save your **First Name**, **Last Name**, **Age**, **Current Weight**, **Height**, and **Gender**.

Additionally on the Profile Screen, **FitFlow** provides the user with the feature of calculating their **Body-Mass-Index (BMI)** by entering their weight and height.

Additionally, you may click the **Log Out** button to be successfully logged out of the program.

(The application currently does not support profile-picture functionality at this time.)

Exiting the Application

At any time, you may click the **Log Out** button to be successfully logged out of the program;

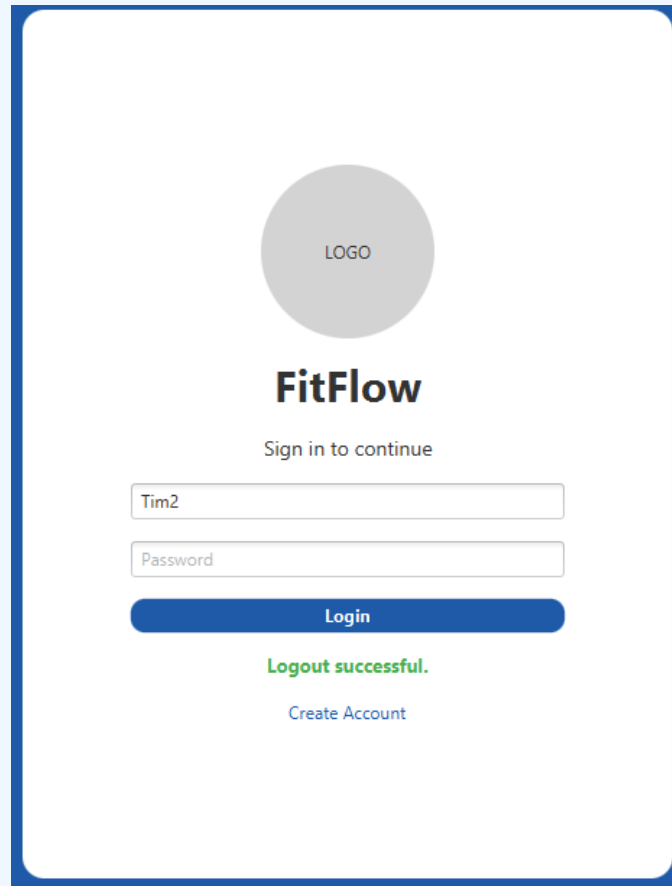


Figure 21. Screenshot of a user successfully logging out of **FitFlow**.

From here, you may successfully exit the application by clicking the **“red x” button** located directly in the upper-right corner of the window, to successfully terminate the program, or alternatively, by terminating it directly from the IDE or the computer’s console terminal.

Troubleshooting Tips

Problem:	Solution:
Program won't start	Verify JavaFX is configured
Can't create account	Passwords must match
Missing images	Verify resources folder exists
Login fails	Check username/password

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Glossary of Terms

General Terms:

- **“BMI”** – “Body mass index (BMI) is a measure of body fat based on height and weight that applies to adult men and women.” (National Heart, Lung, and Blood Institute, 2019)
- **“Dashboard”** – A ‘dashboard’ is a graphical report (as on a website) of various data relevant to a particular business, group, etc. (Merriam-Webster, 2023)

- **“GitHub”** – “GitHub is a cloud-based platform where you can store, share, and work together with others to write code.” (GitHub, 2024)
- **“IDE”** – “An integrated development environment (IDE) is software used by... ..programmers. It packages together various useful developer tools—including a source code editor, build automation tools and a debugger—into a single graphical user interface (GUI) application.” (Schneider & Smalley, 2025)
- **“JavaFX”** – “JavaFX is an open source, next generation client application platform for desktop, mobile and embedded systems built on Java. It is a collaborative effort by many individuals and companies with the goal of producing a modern, efficient, and fully featured toolkit for developing rich client applications.” (Gluon, n.d.)
- **“JavaJDK 26”** – Version 26 of the Java SE platform (Oracle Corporation, 2026).
- **“Login Screen”** – A screen typical of online, mobile, or computer applications where-in a User may enter identifying details, such as a username and password, before the application may display to the User pertinent yet private information.
- **“OS”** – “[S]oftware that controls the operation of a computer and directs the processing of programs (as by assigning storage space in memory and controlling input and output functions)” (Merriam-Webster, 2019a)
- **“Rep”** – “[A] motion or exercise (such as a push-up) that is repeated and usually counted.” (Merriam-Webster, 2026)
- **“Set”** – “[A] group of a specific number of repetitions of a particular exercise.” (Merriam-Webster, 2019b)
- **“User Account”** – An account that stores information about the User to the program.
- **“User Interface”/“UI”** – “Software that is designed to allow a computer user to interact with the operating system of a machine or system (such as by selecting presented options or entering text commands).” (Merriam-Webster, 2020)

FitFlow Specific Terms:

- **“FitFlow”** – ‘FitFlow’ is the name of our application.
- **“Navigation Menu”** – The ‘Navigation Menu’ refers to the icon in the upper right-hand corner that allows the User to navigate to other screens within the FitFlow application.
- **“Profile Screen”** – The ‘Profile Screen’ refers to the screen where the User may enter and record fitness information.

- **“Registration Screen”** – The screen where the User may register an account.
- **“User Profile”** – The account that stores fitness information about the User.
- **“Workout Builder”/“Routine Builder”** – The screen where the User may construct for themselves a workout.
- **“Workout History”** – The screen where the User may review past workouts that were executed upon the program.
- **“Workout Queue”** – The projected timeline that displays to the User during the execution of the workout the current exercise and the upcoming exercises for the selected constructed workout.

Index